

pandas.to_numeric

`pandas.to_numeric()` is used to convert data to a numeric type.

`pd.to_numeric(arg='42.5')` → 42.5

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1st Edition

```
#import Libraries
import pandas as pd
import numpy as np
```

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new concept."

- Practical step By Step Guide With Simple Examples

- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage (pandas Documentation used as reference)

Tips for Effective Use of This Ebook

Use Examples First: If you don't understand the text, go straight to the examples—they're self-explanatory. After reviewing them, return to the text to grasp the practical concepts.

Practice with Datasets: Copy the datasets provided and practice using them to reinforce your understanding.

Click on the blue links to go directly to the section you want to learn about.

Introduction

pandas.to_numeric

`pandas.to_numeric()` is used to convert data to a numeric type.

`pd.to_numeric(arg='42.5')` → 42.5

"pandas.to_numeric" syntax

```
pandas.to_numeric(arg, errors='raise', downcast=None,  
dtype_backend=_NoDefault.no_default)
```

The `arg` parameter

This is the argument to be converted.

The "arg" can be provided as a scalar, a list, a tuple, a 1-dimensional array, or a Series.

When `arg = Scalar`:

A scalar is a single value (like an integer, float, or string) that you want to convert to a numeric type.

```
pd.to_numeric(arg='42.5') → 42.5
```

When `arg = List`:

A list is an ordered collection of elements, such as strings representing numbers, that you want to convert to numeric values.

```
list_values = ['1.0', '2', '3.5']  
pd.to_numeric(arg = list_values) → array([1. , 2. , 3.5])
```

When arg = Tuple:

A tuple is similar to a list but is immutable.

You can pass a tuple of string representations of numbers to be converted.

```
tuple_values = ('4', '5.2', '6')  
pd.to_numeric(arg = tuple_values) → array([4. , 5.2, 6. ])
```

When arg = 1-D Array:

A 1-dimensional array is typically created using a library like NumPy.

You can pass an array of string representations of numbers to convert them.

```
array_values = np.array(['7', '8.8', '9'])  
pd.to_numeric(arg = array_values) → array([7. , 8.8, 9. ])
```

When `arg = Series`:

`Series` is a one-dimensional array-like object in `pandas` that can hold any data type.

You can pass a `Series` of strings to convert them into numeric types.

```
series_values = pd.Series(['10', '11.11', '12'])
pd.to_numeric(arg = series_values)
```

→

0	10.00
1	11.11
2	12.00

dtype: float64

The `errors` parameter

The `"errors"` parameter can be set to `'ignore'`, `'raise'`, or `'coerce'`, with the default being `'raise'`.

`errors='raise'` (Default):

If there's an invalid value that cannot be converted to a numeric type, `pandas.to_numeric()` will raise an exception and stop execution.

```
data = ['10', '20', 'invalid', '30']
pd.to_numeric(data, errors='raise')
```

→

ValueError: Unable to parse string "invalid" at position 2

`errors = 'coerce':`

If there's an invalid

value that cannot be

converted, `pandas.to_numeric()` will

replace it with NaN (Not a Number).

```
data = ['10', '20', 'invalid', '30']
```

```
pd.to_numeric(data, errors='coerce')
```



```
array([10., 20., nan, 30.])
```

`errors='ignore':`

If there's an invalid value,

`pandas.to_numeric()` will

return the input as-is without

converting any data.

```
data = ['10', '20', 'invalid', '30']
```

```
pd.to_numeric(data, errors='coerce')
```



```
array(['10', '20', 'invalid', '30'])
```

The downcast parameter

The parameter "downcast" is used to reduce the size of numerical data types in Pandas to save memory.

The parameter "downcast" is specified as a string, with the default value being None.

Use it when the data's values can be accommodated by a smaller numerical type.

Downcasting Integers:

Original Data: int64 (64-bit integer)

Downcast to: int32, int16, or int8 depending on the range of the values

```
data = pd.Series([1, 2, 3, 4], dtype='int64')
pd.to_numeric(data, downcast='integer')
```



0	1
1	2
2	3
3	4

dtype: int8

Downcasting Unsigned Integers:

Original Data: uint64 (64-bit unsigned integer)

Downcast to: uint32, uint16, or uint8 depending on the range of the values.

```
data = pd.Series([10, 20, 30], dtype='uint64')
pd.to_numeric(data, downcast='unsigned')
```



0	10
1	20
2	30

dtype: uint8

Downcasting Floats:

Original Data: float64 (64-bit floating point)

Downcast to: float32 (32-bit floating point)

```
data = pd.Series([1.1, 2.2, 3.3], dtype='float64')
pd.to_numeric(data, downcast='float')
```



0	1.1
1	2.2
2	3.3

dtype: float32

The `dtype_backend` parameter

The "dtype_Backend" parameter is still experimental.

Back-end data type applied to the resultant DataFrame.

The "dtype_backend" is specified to be either 'numpy_nullable' or 'pyarrow', with 'numpy_nullable' as the default option.

Sources & References

[pandas.to_numeric Documentation](https://pandas.pydata.org/docs/reference/api/pandas.to_numeric.html) (https://pandas.pydata.org/docs/reference/api/pandas.to_numeric.html)

Author Biography

Kichere Magubu is a data enthusiast and content creator.

Other E-books

[Download](https://github.com/kicherethedatascientist/My_e-Books) (https://github.com/kicherethedatascientist/My_e-Books)

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Thank You!

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```
#print("The cell to convert jupyter notebook to html")
#!jupyter nbconvert --to hide_code_html "pandas.to_numeric.ipynb"
```

